

Competitive Programming Codebook

YD

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1 基礎模板

```
#include <bits/stdc++.h>
using namespace std;

int main(){
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    return 0;
}
```

2 資料結構

2.1 Union-Find

```
struct DSU {
    vector<int> p;
    DSU(int n): p(n, -1) {}
    int find(int x){
        return p[x] < 0 ? x : p[x] = find(p[x]);
    }
    void unite(int a, int b){
        a = find(a); b = find(b);
        if(a == b) return;
        if(p[a] > p[b]) swap(a,b);
        p[a] += p[b];
        p[b] = a;
    }
};
```

3 圖論

3.1 Dijkstra

```
vector<long long> dijkstra(int n, vector<vector<pair<
    int,int>>> &g, int s){
    vector<long long> dist(n, 1e18);
    priority_queue<pair<long long,int>, vector<pair<
        long long,int>>, greater<>> pq;

    dist[s] = 0;
    pq.push({0, s});

    while(!pq.empty()){
        auto [d, u] = pq.top(); pq.pop();
        if(d > dist[u]) continue;

        for(auto [v, w] : g[u]){
            if(dist[v] > d + w){
                dist[v] = d + w;
                pq.push({dist[v], v});
            }
        }
    }
    return dist;
}
```

4 數學

4.1 Fast Power

```
int ksm(int x, int y) {
    int ans = 1;
    while (y) {
        if (y & 1)
            ans = ans * x;
        x = x * x;
        y >>= 1;
    }
    return ans;
}

// 矩陣乘法
vector<vector<int>> mul(const vector<vector<int>> &A,
    const vector<vector<int>> &B)
{
    int n = A.size();
    vector<vector<int>> C(n, vector<int>(n, 0));

    for (int i = 0; i < n; i++) {
        for (int k = 0; k < n; k++) if (A[i][k]) {
            for (int j = 0; j < n; j++) {
                C[i][j] = (C[i][j] + A[i][k] * B[k][j]
                    ) % MOD;
            }
        }
    }
    return C;
}

// 矩陣快速冪
vector<vector<int>> ksm(vector<vector<int>> base, int
    exp) {
    int n = base.size();
    vector<vector<int>> res(n, vector<int>(n, 0));

    for (int i = 0; i < n; i++) res[i][i] = 1;

    while (exp > 0) {
        if (exp & 1) res = mul(res, base);
        base = mul(base, base);
        exp >>= 1;
    }
    return res;
}
```

5 字串

5.1 KMP

```
vector<int> build_lps(string s){
    int n = s.size();
    vector<int> lps(n);
    for(int i=1,j=0;i<n;i++){
        while(j>0 && s[i]!=s[j]) j=lps[j-1];
        if(s[i]==s[j]) j++;
        lps[i]=j;
    }
    return lps;
}
```

6 幾何

6.1 Cross Product

```
struct Point {  
    double x, y;
```

```
};
```

```
double cross(Point a, Point b){  
    return a.x * b.y - a.y * b.x;  
}
```